Early mineral entombment as a taphonomic confound in the post-Great Oxidation Event microfossil record

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Abstract

A kinetic calculation establishes that silica entombment of microbial cells is feasible on timescales of hours, shorter than anoxic decay, providing a plausible gate on the preservation of cellular morphology across the Great Oxidation Event (GOE; ∼2.4 Ga). Using the Rimstidt & Barnes (1980) rate law under Precambrian silica saturation ($\Omega \approx 1.8$), a 100 nm silica layer deposits on a microbial cell on the order of hours — an order-of-magnitude feasibility bound rather than a quantitative prediction — comparable to or faster than anoxic decay, though the comparison is uncertain because microbial decay kinetics have not been experimentally constrained. This paper is a Perspective/hypothesis article; it reports no new petrographic, geochemical, or experimental measurements, but offers, alongside the kinetic calculation, a systematic compilation of published preservation data across 15 Precambrian chert-hosted biotas — presented as context rather than as a statistical test — and a specified experimental programme for the definitive test. The compilation is non-independent (both scores derive from the same descriptive literature), so it cannot confirm the mechanism; across 13 biotas of accepted biogenicity, the published descriptions are consistent with, and do not contradict, the hypothesis that early mineral entombment, predominantly silicification, is a dominant taphonomic confound in the apparent diversification of morphologically diagnostic microfossils across the GOE. The hypothesis therefore remains under test; whether any residual GOE-associated variation in fidelity reflects biology or an unmeasured preservational variable remains unresolved.

Keywords: Great Oxidation Event; taphonomic megabias; silicification; iron mineralisation; Archaean–Proterozoic; microfossils; meta-analysis; sterol biosynthesis.

1 Introduction

It has long been recognised that anoxia retards the degradation of sedimentary organic matter, and that the destruction of cellular morphology during early decay proceeds more slowly where molecular oxygen is scarce (Canfield, 1994; Hedges & Keil, 1995). From this premise the Archaean ocean, in which dissolved oxygen was negligible (Pavlov & Kasting, 2002) and sulfate scarce (Habicht et al., 2002; Crowe et al., 2014), might be expected to have favoured the preservation of organic-walled microfossils. The rock record appears to say the opposite: assemblages of taxonomically diagnostic microfossils become abundant and widespread only after the GOE (Javaux & Lepot, 2018; Wacey et al., 2014).

The proposition that an apparent radiation can reflect the opening of a preservational window rather than a biological event is the foundation of the taphonomic-megabias literature. Allison & Briggs (1993) first quantified the non-random, secular distribution of soft-bodied preservation through the Phanerozoic, and Butterfield (2003) argued that the distribution of Burgess-Shale-type preservation constitutes a megabias across the Cambrian explosion. Comparable arguments have been made for phosphatisation windows in the Ediacaran Doushantuo biota and for silicification in the Ediacaran (Tarhan et al., 2016). This framework is long established at the Proterozoic–Cambrian boundary; it has been less developed across the GOE.

The contribution of the present paper is a systematic meta-analysis of the published petrographic record, framed as an explicit hypothesis under test. As stated in the abstract, this is a Perspective/hypothesis article reporting no new measurements; its four contributions are a compilation, a meta-analytical test with sensitivity checks, a kinetic feasibility calculation, and a specified experimental programme. We compile silicification and cellular-fidelity scores for 15 Precambrian chert-hosted microfossil biotas and ask two questions: does silicification predict cellular fidelity across the record, and does GOE position retain explanatory power once silicification is controlled for? A kinetic calculation then assesses whether catalysed silica entombment can outpace decay. The compilation is semi-quantitative and drawn from a literature dominated by a few research groups, a limitation stated explicitly below.

2 Background

2.1 Redox across the GOE

The disappearance of mass-independent sulfur isotope fractionation between ∼2.45 and 2.32 Ga marks the GOE (Holland, 2006; Bekker et al., 2004; Pavlov & Kasting, 2002; Farquhar et al., 2000); the transition was protracted and regionally diachronous (Lyons et al., 2014; Philippot et al., 2018). Iron speciation indicates anoxic, ferruginous deposition throughout the Archaean and much of the Proterozoic (Canfield, 1998; Poulton & Canfield, 2005, 2011), euxinia being spatially restricted (Reinhard et al., 2009). Seawater sulfate rose across the GOE (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014), but sulfur isotope evidence of this rise extends into the Neoarchean (Reinhard et al., 2009), and signals of oxidative weathering precede the GOE in several basins (Anbar et al., 2007).

2.2 The microfossil record

The Archaean record of cellular morphology is sparse and contested. The ∼3.46 Ga Apex chert filaments (Schopf, 1993) were reinterpreted as abiogenic (Brasier et al., 2002), and although better-preserved Archaean microbiotas are known from the ∼3.4 Ga Strelley Pool Formation (Sugitani et al., 2010; Delarue et al., 2020), their cellular fidelity is modest. The ∼2.45–2.21 Ga Turee Creek Group spans the GOE and preserves diverse microfossil communities in black chert (Barlow & Van Kranendonk, 2018), the taphonomy and iron mineralisation of which are documented by Lepot et al. (2017b). Gunflint-type biotas of ∼1.9 Ga age carry filaments, spheroids and spindle-shaped forms in cellular detail (Lepot et al., 2017a; Javaux & Lepot, 2018).

2.3 The silica cycle

Before the evolution of silica-biomineralising organisms the ocean was saturated with respect to amorphous silica, with dissolved concentrations of order ∼2 mM and abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2007). The capacity for early silica precipitation was therefore available on both sides of the GOE; silica was not drawn down until the much later radiation of sponges, radiolaria and diatoms.

2.4 A biochemical oxygen constraint on eukaryotes

Sterol biosynthesis is obligately aerobic: the cyclisation and oxidative modification of squalene requires molecular oxygen, on the order of eleven molecules of O_2 per sterol (Gold et al., 2017; Desmond & Gribaldo, 2009). Crown eukaryotes synthesise sterols, and the last eukaryotic common ancestor is inferred to have done so (Desmond & Gribaldo, 2009). This provides a mechanistically independent reason to expect crown-eukaryote diversification to track the availability of oxygen, though molecular-clock estimates of when that diversification occurred are themselves partly calibrated to GOE geochemistry (Gold et al., 2017), so the independence that holds is biochemical rather than chronological.

3 Material and approach

Fifteen chert-hosted microfossil biotas were compiled (Table 1): thirteen of accepted biogenicity and two (Apex, Dresser) of contested biogenicity. The sample spans the Archaean to the Palaeoproterozoic, from the ∼3.48 Ga Dresser Formation to the ∼1.80 Ga Duck Creek Formation, and includes pre-GOE, GOE-spanning, and post-GOE units. Occurrences and their sources were drawn from the reviews of Javaux & Lepot (2018) and Wacey et al. (2014), supplemented by primary petrographic studies where available. No new petrographic measurements are reported; the new contribution is the systematic compilation and its statistical analysis.

For each biota two scores were assigned on 0–3 ordinal scales. *Silicification* (0–3) was coded from the timing and fabric of quartz cementation — pre-compaction, wall-conforming microquartz or chalcedony scoring highest; late megaquartz void-fill scoring lowest — using petrographic criteria kept logically independent of the fidelity outcome. *Cellular fidelity* (0–3) was coded from the degree of cellular detail preserved: 0, no cellular structure; 1, mat or globule textures only; 2, partial cellular detail; 3, diverse assemblages with well-preserved cellular contents. The scoring criteria, and the unit to which each applies, are given in Table 1.

The independence of the silicification coding from the fidelity outcome is logical rather than source-based: the petrographic and fidelity assessments alike are drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey). Correlations were estimated with the Spearman rank coefficient. The implications of this non-independence for interpreting the statistics are taken up in the Discussion (Limitations).

Table 1: Systematic compilation of 15 Precambrian chert-hosted microfossil biotas. Silicification score (0–3) is coded from quartz cement fabric and timing, independent of the fidelity outcome. Cellular fidelity score (0–3) is coded from the degree of cellular detail preserved. GOE position: pre, spans, or post. Crosses mark contested biogenicity.

Formation (age; GOE)	Biogenicity	Silic.	Fid.	Source
Strelley Pool (∼3.4 Ga; pre)	Accepted	3	2	(Sugitani et al., 2010; Delarue et al., 2020; Wacey et al., 2012)
Kromberg Fm (∼3.42 Ga; pre)	Partial (mats)	2	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Josefdal Chert (∼3.33 Ga; pre)	Partial (mats)	2	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Moodies Gp (∼3.22 Ga; pre)	Accepted (mats)	1	1	(Homann et al., 2015)
Farrel Quartzite (∼3.0 Ga; pre)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
Tumbiana Fm (∼2.72 Ga; pre)	Accepted (globules)	1	1	(Lepot et al., 2009; Decraene et al., 2021)
Belingwe GSB (∼2.70 Ga; pre)	Accepted (mats)	1	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Turee Creek Gp (∼2.33 Ga; spans)	Accepted	3	3	(Lepot et al., 2017b; Barlow & Van Kranendonk, 2018)
Belcher Gp (∼2.0 Ga; post)	Accepted	3	3	(Javaux & Lepot, 2018)
Gunflint Fm (∼1.88 Ga; post)	Accepted	3	3	(Lepot et al., 2017a; Wacey et al., 2013; Alleon et al., 2016a)
Sokoman IF (∼1.88 Ga; post)	Accepted	3	3	(Javaux & Lepot, 2018)
Duck Creek Fm (∼1.80 Ga; post)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
Rove Fm (∼1.85 Ga; post)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
† Apex chert (∼3.46 Ga; pre)	CONTESTED	2	1	(Schopf, 1993; Brasier et al., 2002; Wacey et al., 2014)
† Dresser Fm (∼3.48 Ga; pre)	CONTESTED	1	0	(Wacey et al., 2014)

† Contested biogenicity; included for sensitivity but excluded from the primary analysis.

The raw scores are presented in Table 1 rather than as a scatter plot, to avoid implying a statistical strength the non-independent compilation cannot support.

4 Results

The kinetic calculation is the spine of the argument and is presented first. The compilation (Table 1) follows as a structured summary of the published record, presented as context rather than as a statistical test.

4.1 Kinetic feasibility of early entombment

The feasibility of early entombment can be assessed from the rate law for amorphous silica precipitation. For amorphous silica at 25°C, the forward rate constant is $k \approx 1.0 \times 10^{-10}$ mol cm^{-2} s^{-1} (Rimstidt & Barnes, 1980). The net precipitation rate per unit surface area is $R = k(\Omega - 1)$, where $\Omega = C/C_{\text{eq}}$ is the saturation ratio. For the Precambrian ocean, with dissolved silica $C \approx 2$ mM (Siever, 1992) and amorphous silica solubility $C_{\text{eq}} \approx 1.1$ mM, $\Omega \approx 1.8$ and $(\Omega - 1) \approx 0.8$. For a spherical cell of 1 μm radius (surface area $A \approx 1.26 \times 10^{-7}$ cm^2), the precipitation rate is approximately 1.0×10^{-17} mol s^{-1}, and the time to deposit a 100 nm silica layer (requiring $\approx 4.6 \times 10^{-14}$ mol SiO_2) is ≈ 1.3 hours. This is the timescale against which decay must be judged. The decay half-life of cellular ultrastructure under anoxic conditions is taken from experimental taphonomy of macroscopic metazoans (days to weeks; Corthésy et al., 2025), but the decay kinetics of micron-scale prokaryotic cells, which have substantially higher surface-area-to-volume ratios and different wall chemistry, may be faster. The entombment–decay race is therefore likely competitive for microbial cells, not the comfortable one-to-two-order-of-magnitude advantage the macro-animal benchmark implies. This near-competitiveness is itself consistent with the facies-dependent variability in preservation quality observed across Precambrian cherts: whether entombment wins depends on the precise balance of local supersaturation, catalytic surface availability, and microbial decay rate. The calculation addresses the growth rate of an established silica layer,

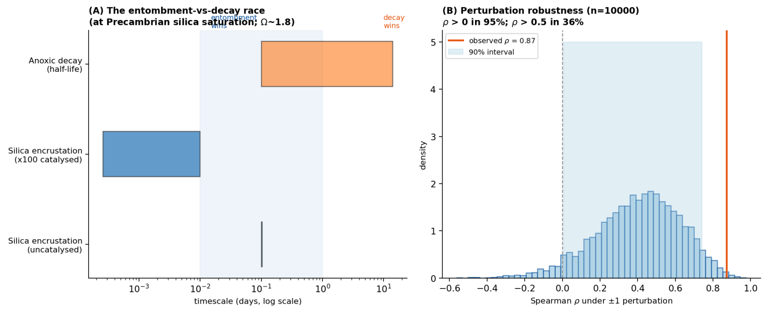

Figure 1: The entombment–decay race at Precambrian silica saturation ($\Omega \approx 1.8$). Abiotic silica encrustation (Rimstidt & Barnes, 1980) rate law deposits a 100 nm layer in $\approx$1 hour, comparable to or faster than microbial anoxic decay over days to weeks, though the latter is referenced to macro-animal experiments (Corthésy et al., 2025) and is uncertain for prokaryotic cells. Surface catalysis or inhibition introduces large uncertainty in the effective rate constant. The inset perturbation histogram addresses coding noise but not the shared-provenance circularity discussed in the text.

not the nucleation induction time on an organic surface, which in real silica systems can be the rate-limiting step; if nucleation induction is slow (hours to days), the effective entombment time increases accordingly. The kinetic calculation therefore establishes that entombment is *feasible* on the timescale required, but the outcome of the entombment–decay race depends critically on the surface chemistry of the cells, the nucleation induction time, and the degree of local supersaturation. The effective rate constant on microbial surfaces is uncertain by orders of magnitude: cell wall functional groups catalyse silica nucleation (enhancing the rate), while organic coatings can inhibit precipitation. This sensitivity to surface properties, which varies with organism and environment, provides a mechanistic explanation for the facies-dependent variability in preservation quality across Precambrian cherts (Fig. 1). Experimental work confirms that early entombment within silica minimises the molecular degradation of microorganisms during advanced diagenesis (Alleon et al., 2016b), providing direct support for the timing effect the hypothesis predicts.

The result is sensitive to the assumed solubility of amorphous silica (C_{eq}). Published values range from approximately 1.1 to 1.8 mM at 25°C. At $C_{\mathrm{eq}} = 1.1$ mM the calculation gives $\sim$1.3 hours; at $C_{\mathrm{eq}} = 1.5$ mM, $\sim$3 hours; at $C_{\mathrm{eq}} = 1.8$ mM, $\sim$10 hours. In all geologically plausible cases ($C_{\mathrm{eq}} < C$, necessary for chert deposition), the entombment timescale remains shorter than the anoxic decay half-life (days to weeks). Only at $C_{\mathrm{eq}} = C$ would supersaturation vanish entirely, but this would preclude chert precipitation altogether and is inconsistent with the abundant Precambrian chert record.

4.2 The compilation as descriptive context

The compilation (Table 1) is presented as a structured summary of the published record rather than as a statistical test. Read descriptively, silicification and fidelity scores move together in direction across the 13 accepted-biogenicity biotas (Spearman $\rho = 0.87$), and a permutation test (100,000 random reassignments) indicates this is stronger than expected by chance. The robustness checks (permutation, null-model, perturbation) address sampling noise and coding subjectivity; the perturbation analysis (Fig. 1B) was generated by applying $\pm$1 random noise to each of the 26 Table 1 scores over 10,000 trials, using a deterministic random seed. Including the two contested-biogenicity biotas (Apex, Dresser; $n = 15$) leaves the direction of association unchanged. The interpretive limits of this non-independent compilation are stated in full in the Discussion (Limitations).

A partition by research group offers a partial test of one robustness concern. Restricting the compilation to biotas whose preservation is described principally by the Lepot group (Strelley Pool, Tumbiana, Turee Creek, Gunflint; $n = 4$), the association remains positive ($\rho = 0.82$). Restricting to biotas described principally by other groups ($n = 9$), the association is similarly positive ($\rho = 0.83$, $p = 0.006$). The partition tests whether one group's descriptive conventions drive the result, not whether silicification and fidelity scores can be independently ascertained from the same source text — which is the deeper problem the robustness checks cannot address.

5 Within-formation evidence: the Gunflint iron covariation

Within the Gunflint biota, morphospecies retaining cellular contents carry 3×10^8–10^9 intracellular Fe atoms $\mu\mathrm{m}^{-3}$ while empty sheaths remain at chert background ($\sim 1.5 \times 10^6$ atoms $\mu\mathrm{m}^{-3}$; Fig. 2) (Lepot et al., 2017a). This covariation links fidelity to early mineralisation but adds little probative weight beyond the cross-formation compilation: Lepot et al. (2017a) interpret the iron as consistent with in vivo biomineralisation

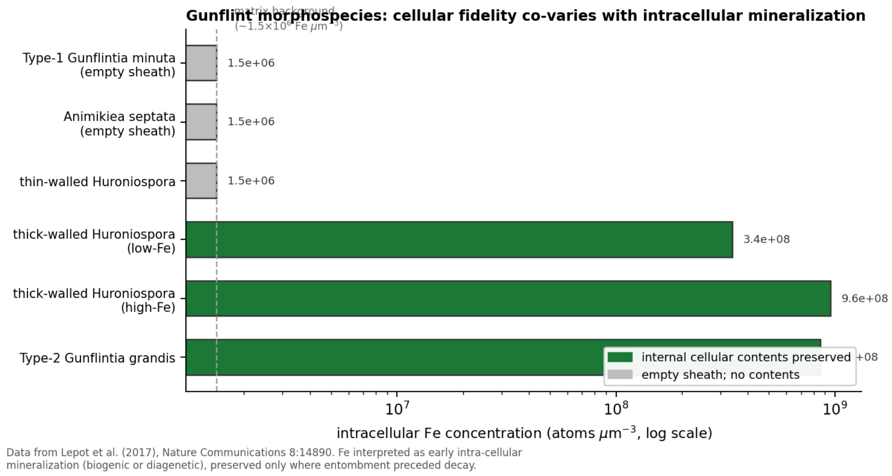

Figure 2: Intracellular Fe concentration in Gunflint morphospecies, against the preservation of cellular contents. The covariation links fidelity to early mineralisation but is confounded by possible in vivo biomineralisation and wall-recalcitrance effects (Lepot et al., 2017a).

rather than a purely external precipitate, and the cellular morphospecies are thick-walled forms in which wall recalcitrance co-varies with morphotype rather than with entombment timing.

5.1 Why redox and the silica cycle do not supply a clean GOE control

Iron speciation indicates pervasive ferruginous deposition on both sides of the GOE (Canfield, 1998; Poulton & Canfield, 2011), and the sulfur isotope signal of rising sulfate extends into the Neoarchean (Reinhard et al., 2009; Philippot et al., 2018). The sulfate rise bears on microbial sulfate reduction and the sulphurisation of organic matter, affecting the survival of bulk kerogen rather than the templating of cellular morphology. The capacity for early silica precipitation was likewise unchanging across the GOE (Siever, 1992; Perry & Lefticariu, 2007). Any residual GOE-associated variation in fidelity is therefore not readily attributed to any single measured redox or silica variable, and its origin remains open.

6 A case that cannot adjudicate: pre-GOE silicified cherts

The most promising candidate for a sharp test of the silicification hypothesis would be a pre-GOE chert that is demonstrably early-silicified. The ~3.46 Ga Apex chert and the ~3.48 Ga Dresser Formation are pre-GOE, silicified, and rich in cell-like carbonaceous structures, and so might appear to provide exactly such a test. They cannot. The Apex filaments described as microfossils (Schopf, 1993) were reinterpreted as abiogenic mineral artefacts (Brasier et al., 2002), and high-resolution reinvestigation has reinforced an abiogenic, hydrothermal origin for many such structures (Wacey et al., 2014). Where biogenicity is itself contested, no observation of cellular fidelity can adjudicate the entombment-timing hypothesis, because the question of whether these structures were ever cells is logically upstream of the question of how faithfully their morphology was preserved. A chert of disputed biogenicity is therefore structurally incapable of testing the hypothesis, and we treat Apex and Dresser as informative about the nature of silicified cherts rather than as a falsification test that the hypothesis has survived.

The observation these cherts do support is narrower but worth stating plainly. Silicification preserves morphology indiscriminately — whether the substrate is biological or abiotic — so silicified cherts will contain cell-like structures of mixed origin, and the biogenicity question is upstream of the fidelity question. The scarcity of unambiguous pre-GOE microfossils is consequently not evidence that silicification was feeble in the Archaean (it demonstrably was not (Siever, 1992; Perry & Lefticariu, 2007)), but that silicification records both biology and abiotic look-alikes, leaving few structures that meet modern biogenicity criteria.

7 Origins, first appearances, and the sterol constraint

Molecular clocks place the stem origin of cyanobacteria, and oxygenic photosynthesis, in the Archaean (Fournier et al., 2021), whereas the crown-group diversification of extant Oxyphotobacteria postdates the rise of oxygen (Shih et al., 2017). Molecular clock estimates for the last eukaryotic common ancestor are model-dependent but several precede the first widely accepted eukaryotic microfossils (Betts et al., 2018; Eme et al., 2014)

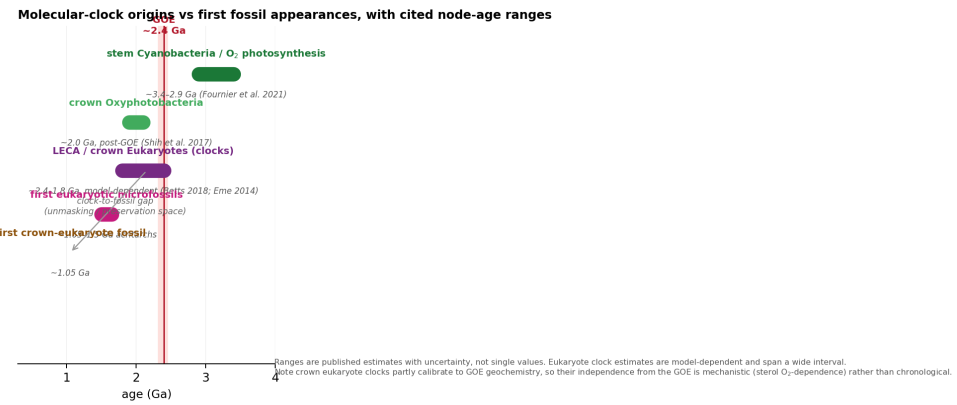

Figure 3: Molecular clock estimates of lineage origin against first fossil appearances, with published node-age ranges. The stem origin of cyanobacteria lies in the Archaean (Fournier et al., 2021); crown Oxyphotobacteria diversify around or after the GOE (Shih et al., 2017); eukaryote estimates are model-dependent (Betts et al., 2018; Eme et al., 2014) and partly calibrated to GOE geochemistry.

(Fig. 3). The biomarker record cannot resolve the resulting gap: the steranes and 2-methylhopanes reported from ∼2.7 Ga shales (Brocks et al., 1999) reside in younger carbonate veins (Rasmussen et al., 2008) and are indistinguishable from procedural blanks in ultraclean cores (French et al., 2015).

The obligate aerobicity of sterol synthesis (Gold et al., 2017; Desmond & Gribaldo, 2009) cuts across a purely preservational reading for eukaryotes: crown eukaryotes have a mechanistic reason to diversify once oxygen became available, regardless of preservation. The clock-to-fossil gap for eukaryotes is thus not necessarily a preservational lag and may reflect genuine, oxygen-limited diversification. This supplies one candidate biological interpretation of the GOE-associated variation discussed in the Results and Discussion: if eukaryote and cyanobacterial diversification were oxygen-limited, the post-GOE rise in fidelity could partly record real biological change rather than preservation alone. The unmasking hypothesis is consequently better supported for cyanobacteria, whose metabolism clocks to the Archaean, than for sterol-synthesising eukaryotes, and any reading must be lineage-conditional.

8 Discussion

The kinetic calculation shows that silica entombment is feasible on the timescale required to gate cellular preservation, and the compilation is consistent with — but cannot confirm — the hypothesis that silicification is a dominant predictor of cellular fidelity across the Precambrian microfossil record. GOE position may retain some explanatory power not captured by the silicification score, though the magnitude cannot be resolved at this sample size. The defensible position is therefore narrower than a claim that silicification alone explains the post-GOE pattern. The preservation of cellular morphology in Precambrian cherts is conditioned by early mineralisation, predominantly silicification, that operated throughout but was expressed unevenly, depending on depositional facies and the timing of cementation relative to decay. Where entombment was early and complete, morphology survives; where it was weak, kerogen and mats remain. Oxygenation plausibly modified organic-matter decay and kerogen sulphurisation (Lepot et al., 2009; Decraene et al., 2021) but did not gate the silicification on which cellular fidelity primarily depends. The origin of any such GOE-associated variation — whether biological diversification, an unmeasured preservational change, or sampling bias — cannot be resolved from this compilation alone.

Silicification is not the only mineral-entombment pathway capable of preserving cellular morphology. Phosphatisation preserves subcellular ultrastructure in Neoproterozoic phosphorites (the Doushantuo biota; Tarhan et al., 2016) and requires phosphate-rich porewaters that may have been more locally available after the GOE. Clay-mineral templating enhances cellular preservation in Proterozoic lacustrine deposits (Wacey et al., 2014). The hypothesis advanced here — that mineral entombment gates morphological fidelity — applies to these pathways in principle; each has its own temporal and environmental distribution, and a complete account of preservational megabias across the GOE would need to weigh all three rather than silicification alone.

The post-GOE signal is likely a mixture of taphonomic unmasking and genuine biological diversification, and several non-taphonomic mechanisms could contribute to a real increase in preservable morphology that the hypothesis advanced here does not exclude. Increased cell-wall robustness, driven by adaptation to a more oxidising atmosphere or to UV stress, would enhance preservation potential independently of silicification — a biological change that mimics a preservational signal. The rise of predation and grazing could drive morphological diversification through defensive or competitive innovations. More complex mat architectures, with layered communities and differentiated niches, would produce a greater volume and variety of preservable biomass. The present paper argues only that the taphonomic component is non-negligible and specifies how the partition could be made.

8.1 Limitations

The silicification and fidelity scores are ordinal and were assigned from the descriptive literature; they are not source-independent, because the petrographic and fidelity assessments alike are drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey). The robustness checks confirm that the observed association is stable to random coding noise, but they cannot exclude the possibility that both scores are partly labels for the same underlying descriptive judgement, or that the association partly reflects shared descriptive provenance rather than two independently measured properties. The statistics should not, therefore, be read as a confirmatory test. Only blind, cross-group rescoring — in which silicification is coded from mineralogical observations and fidelity from palaeobiological observations, by separate research groups unaware of each other's scores — can resolve this. None of these limitations undermines the principal observation that the silicification–fidelity association is robust in direction, but they bound the strength of the claim about the residual GOE signal, whose biological versus preservational origin the compilation cannot decide.

There is a more fundamental concern than shared provenance, which the group-partition check cannot resolve. Cellular fidelity is, in the nature of chert-hosted assemblages, observable only where a mineral phase has held cellular structure in place before decay destroyed it. Where silicification was absent or late, cells were not preserved — and therefore cannot be scored for fidelity. The compilation therefore selects, in part, on the dependent variable: formations with poor silicification appear in the compilation with fidelity scores of 0–1 not because their cells were intrinsically unpreservable, but because without silica they were not preserved at all. The correlation between silicification and fidelity is thus partly definitional rather than purely empirical. This does not invalidate the hypothesis — it is the mechanism the hypothesis proposes — but it means the compilation cannot independently test the mechanism; it can only document that the pattern is consistent with it.

One confound the compilation controls for by design is thermal maturity. All 13 formations are of low metamorphic grade, ranging from sub-greenschist to prehnite–pumpellyite facies, so thermal maturity has effectively no variance across the sample and cannot account for the observed range in cellular fidelity. This control-by-design is a strength, though it also means the analysis cannot assess whether the hypothesis holds among higher-grade successions.

The Ediacaran silicification dataset of Tarhan et al. (2016) offers a potential template for a more statistically powerful version of this compilation: the Ediacaran–Cambrian boundary has more well-characterised biotas, more independent research groups, and denser temporal sampling than the Archaean–Proterozoic. Extending the megabias framework developed here to that boundary, where silicification is already implicated in exceptional preservation, would provide a stronger test.

The implications are qualified accordingly. Abundance and diversity counts across the GOE cannot be read as proxies for biomass without facies correction. The clock-to-fossil gap is consistent with a preservational lag but, for sterol-synthesising lineages, may equally reflect oxygen-limited diversification — a reading consistent with the GOE-associated variation noted above. For the search for ancient biosignatures, the corollary is that the preservation of morphology requires a silica-precipitating facies rather than the presence of oxygen, but the appearance of morphological diversity after the GOE should not be attributed to preservation alone.

8.2 What would be needed: an experimental and analytical programme

The hypothesis is testable; we specify three approaches in order of decisiveness.

(a) Controlled silicification–decay experiments. The confound that most limits the within-Gunflint evidence is the co-variation of wall recalcitrance and mineral-nucleation potential with morphotype. A laboratory programme can hold morphotype constant and vary only the timing of silica addition. Microbial cultures spanning thick- and thin-walled morphotypes (e.g. cyanobacteria and other bacteria) would be incubated under factorial $O_2 \times$ sulfate $\times$ Fe conditions, with controlled silica added at defined intervals after death. Blind-scored cellular fidelity, silica nucleation timing, and intracellular mineralisation would then reveal whether entombment timing — rather than wall recalcitrance — is the causal variable. The experimental demonstration that early silica entombment minimises molecular degradation during advanced diagenesis (Alleon et al., 2016b) provides

the mechanistic foundation for such a programme; what is needed is the controlled timing manipulation that no published experiment has yet performed.

(b) A blind, cross-group within-formation regression. This is the test the present compilation proposes but does not perform at the population level. Cellular fidelity and a quantitative silicification index would be scored independently by different research groups, blind to one another's scores, on 30–50 microfossil populations per formation, and regressed while controlling for thermal maturity (Raman R_2). A null relationship across formations of comparable grade would weigh against the hypothesis; a consistent positive relationship would support it.

(c) A systematic survey for the falsifier. A single pre-GOE chert (or indeed any-age chert) carrying microfossils of undisputed biogenicity, demonstrably early silicification, and poor cellular fidelity would falsify the hypothesis outright. A targeted search of curated chert collections, screened first for biogenicity and silicification timing and then scored blind for fidelity, is the most economical route to a decisive test. No such case is currently known.

9 Conclusions

The kinetic calculation is the paper's strongest result. Using the Rimstidt & Barnes (1980) rate law under Precambrian silica saturation ($\Omega \approx 1.8$), abiotic silica precipitates a 100 nm entombment layer in $\approx$1.3 hours — comparable to or faster than anoxic decay over days to weeks (Corthésy et al., 2025), though microbial decay kinetics are referenced to macro-animal experiments and are uncertain for prokaryotic cells — establishing that early mineral entombment, predominantly silicification, is feasible on the timescale required to gate the preservation of cellular morphology across the GOE. The systematic compilation of 15 Precambrian chert-hosted biotas is consistent with this hypothesis: across 13 biotas of accepted biogenicity, the published descriptions of silicification and cellular fidelity move together in direction, though the compilation is non-independent and cannot confirm the mechanism (Limitations). Whether any residual GOE-associated variation in fidelity reflects genuine biological diversification, an unmeasured preservational change, or sampling bias remains unresolved by the compilation. The hypothesis would be falsified by (i) a pre-GOE chert bearing confirmed genuine microfossils with demonstrably early silicification and poor cellular fidelity, or (ii) a within-formation regression showing no relationship between fidelity and a quantitative silicification index scored blind and across research groups; neither is known, and the second is achievable on existing material. The experimental and analytical programme needed to pursue these tests is specified above. Confirmation awaits that programme; the present paper makes the case that the hypothesis is worth testing.

Data availability

All data are compiled from previously published sources cited in Table 1. The compilation scores (Table 1), the permutation and perturbation analysis routines (Python), and the kinetic calculation code are provided as electronic supplementary material.

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